

Mobile Networks and IoT 2026/2027

PL Class 01 – Creating your first IoT program

Vasco Pereira (vasco@dei.uc.pt)

Dep. Eng. Informática da Faculdade de Ciências e Tecnologia da Universidade de Coimbra

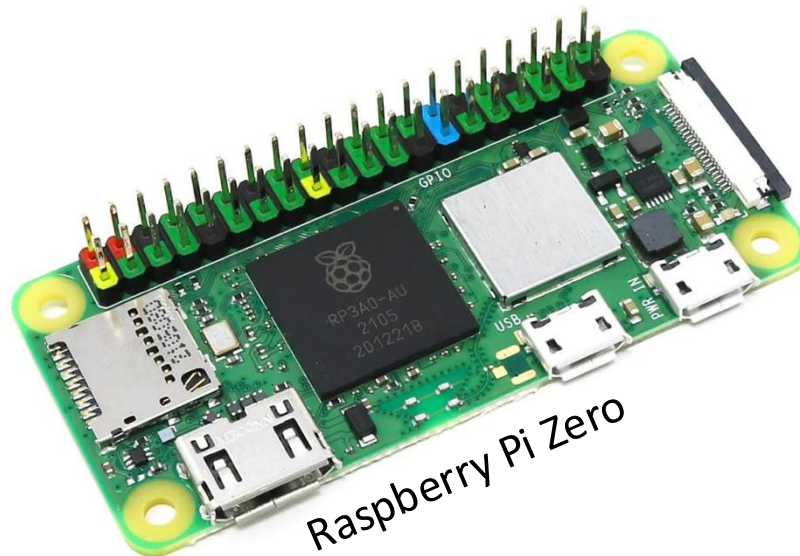
Raspberry Pi introduction

Mobile Networks and Internet of Things

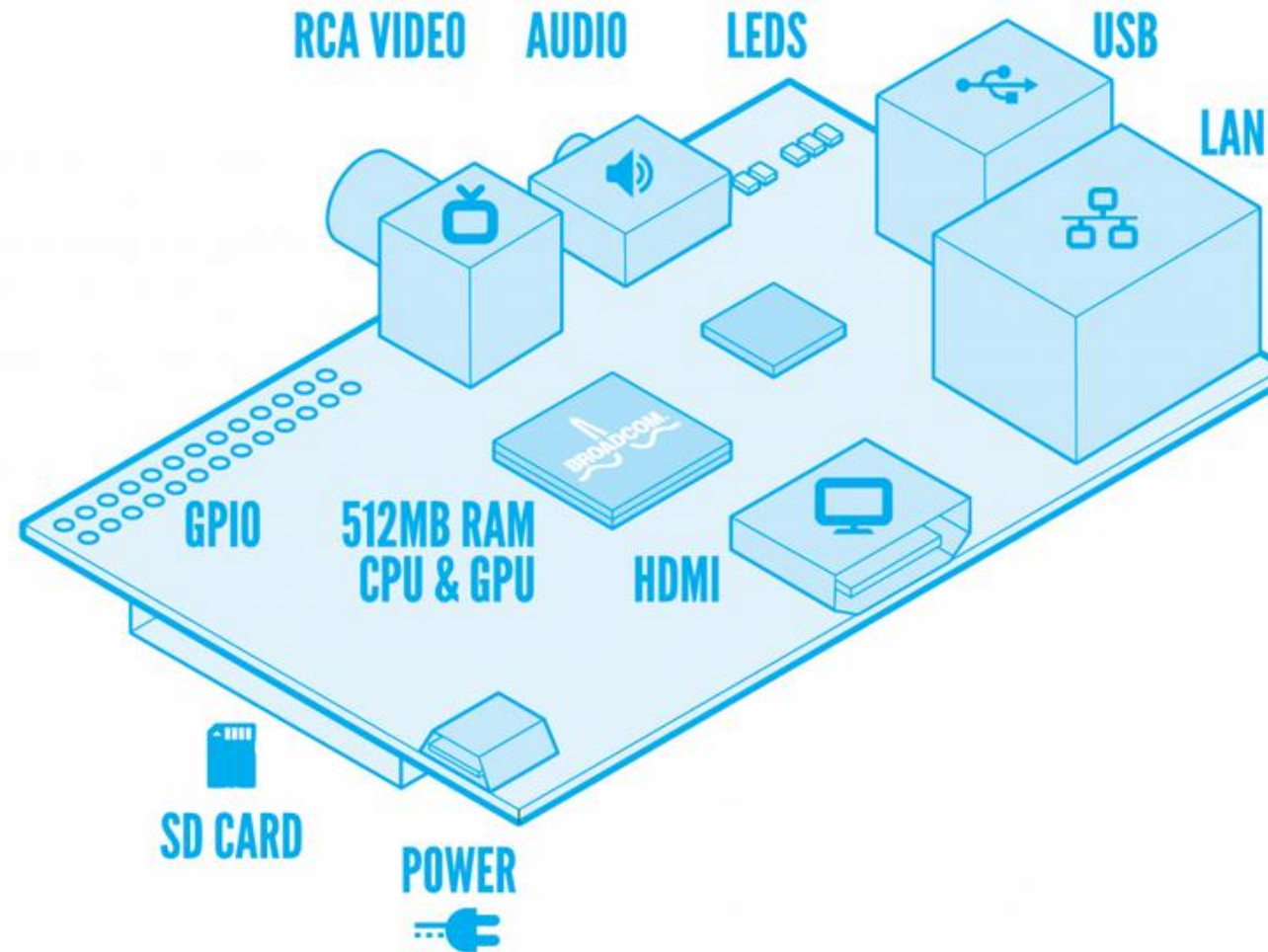
Raspberry Pi

3

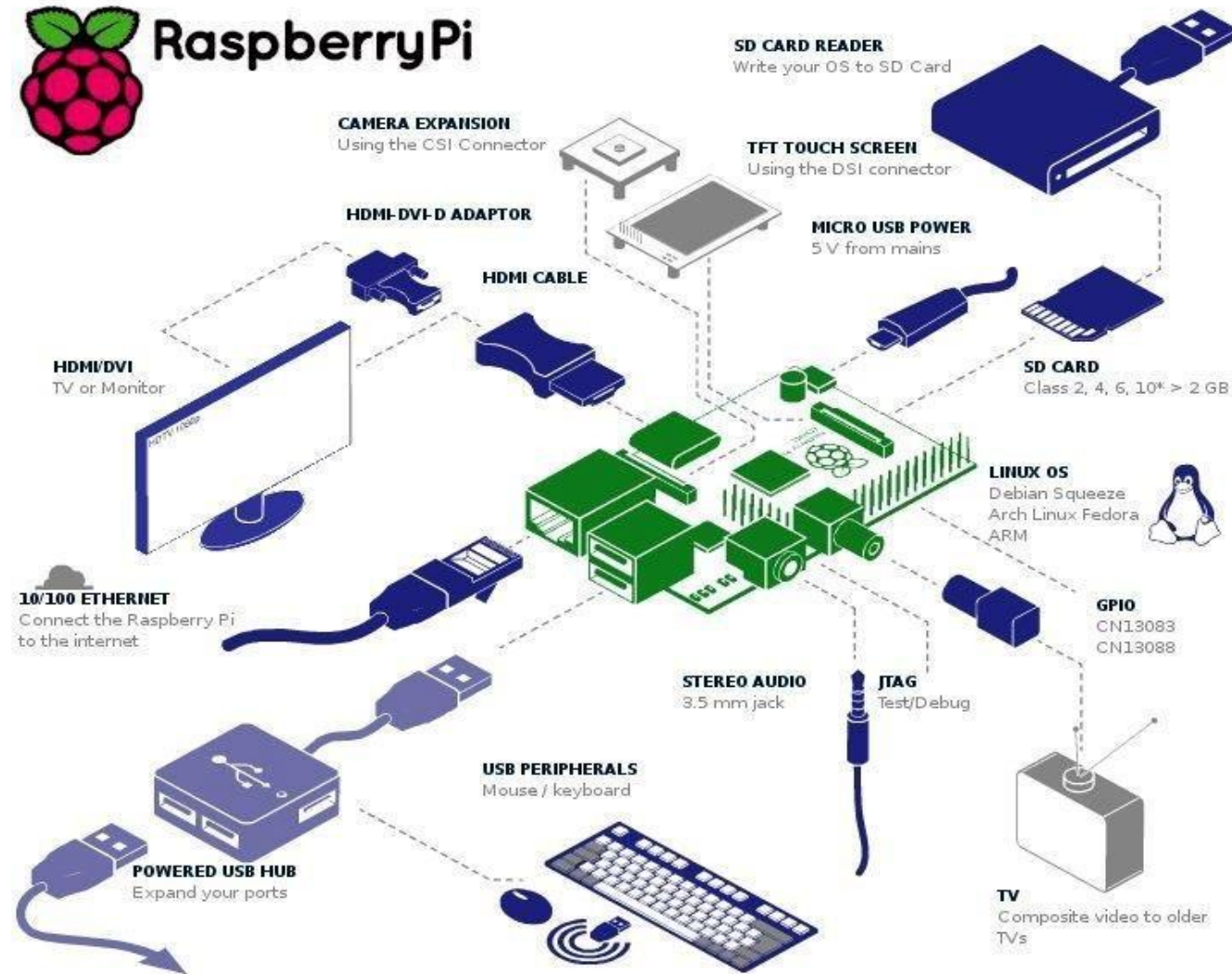
- Tiny, single-board, affordable computer;
- Teaching basic computer science;
- Currently is used for many purposes including weather monitoring and home and industrial automation.



Raspberry Pi



Raspberry Pi



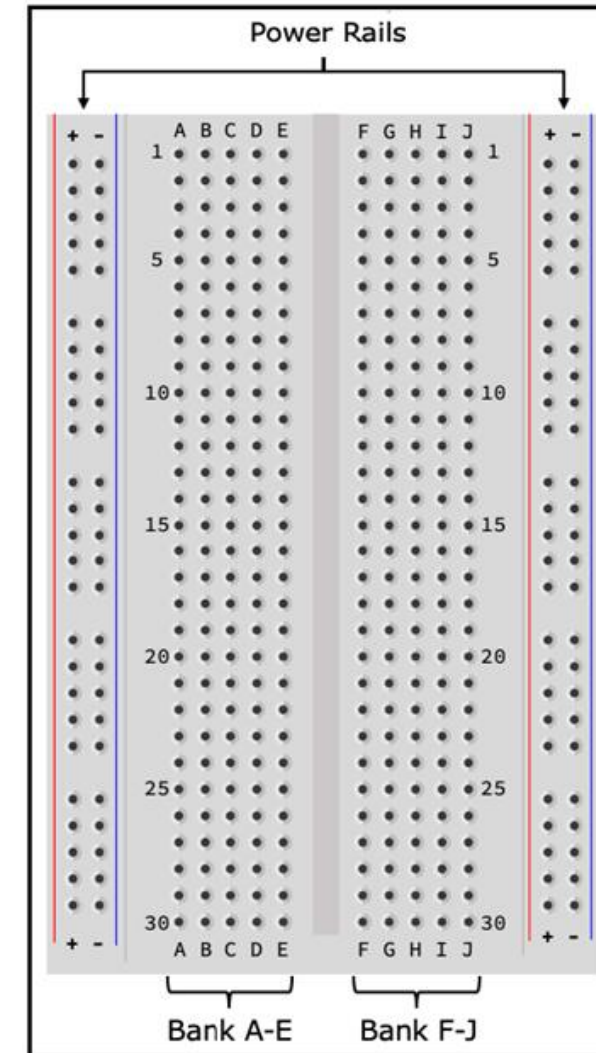
Breadboard

Mobile Networks and Internet of Things

Breadboard

7

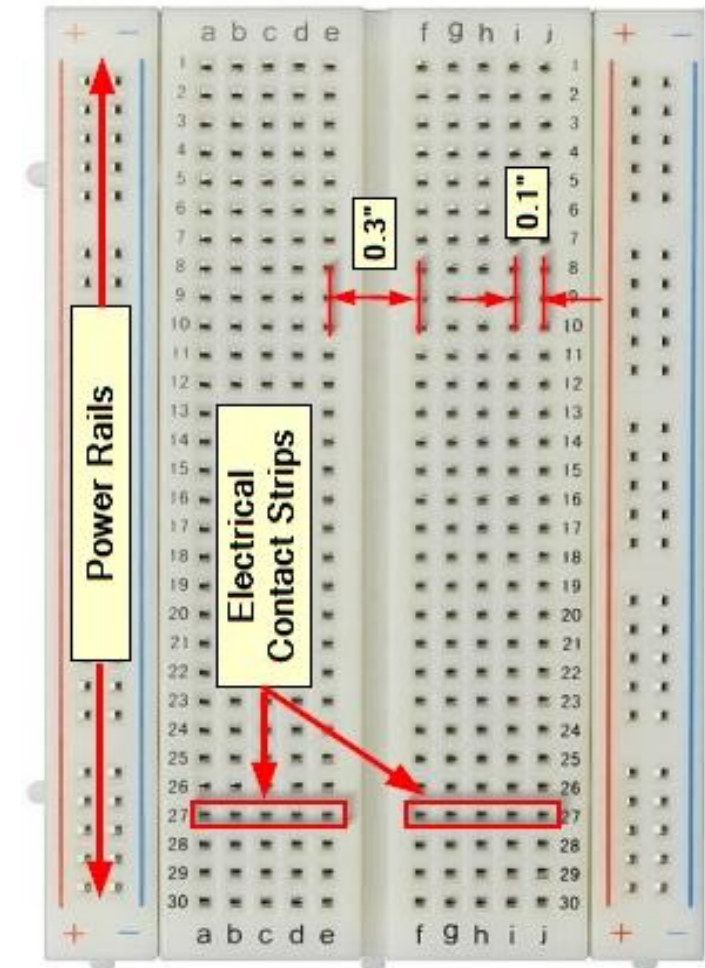
- An electronic breadboard (or protoboard) is a prototyping board that helps to electrically connect components and wires quickly and easily.



Breadboard

8

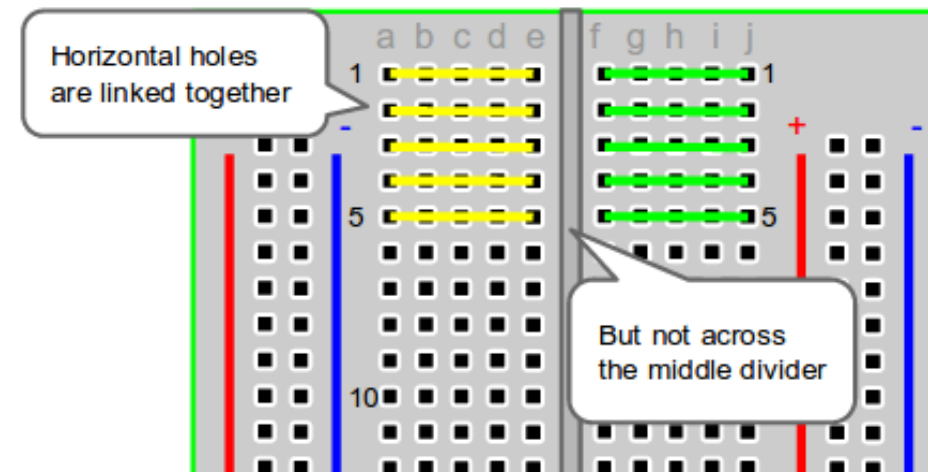
- The holes in the breadboard are where electrical components and wires are placed to electrically connect them. The holes are electrically connected;
- The two outer columns of holes are commonly referred to as *power rails*. There is a positive (+) column and a negative (-) column on either side of the breadboard;
- They do not provide power themselves. They need a power source such as a power supply or battery connected to them to provide power.



Breadboard

9

- The centre of the breadboard has two banks of holes, which have been labelled Bank A-E and Bank F-J. Each row of holes in a bank is electrically connected. For example, holes A1 through to E1 are electrically connected, as are holes F1 through to J1. However, A1-E1 are not electrically connected to F1-J1 because they are on a separate bank.



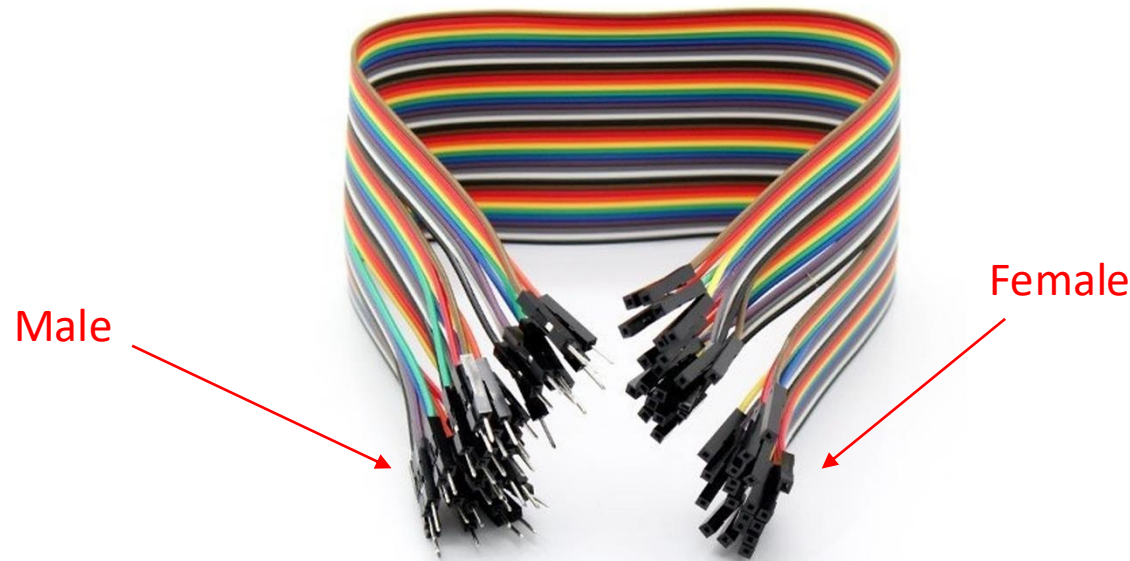
Jumper Wires

Mobile Networks and Internet of Things

Jumper Wires

11

- Jumper wires are used to make it easier to manage circuits built on a breadboard;
- The wires are usually cut to lengths such that they fit neatly between two holes in the breadboard, but it is easy enough to create your own using a length of wire and some wire strippers;



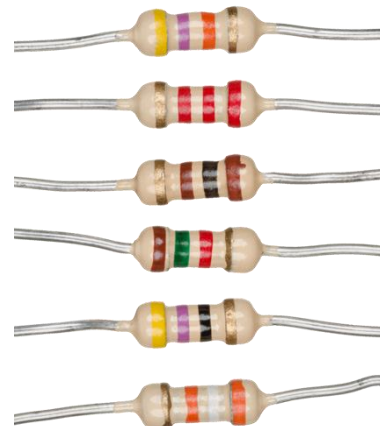
Resistors

Mobile Networks and Internet of Things

Resistors

13

- A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element;
- Used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses;
- Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment.



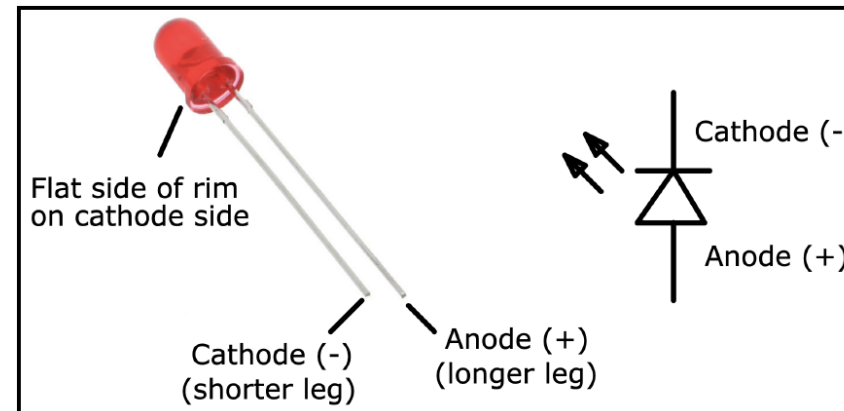
LED Light

Mobile Networks and Internet of Things

Led Light

15

- A LED (*Light-Emitting Diode*) is a small, yet bright, light made of a tiny crystal that emits a colour when electricity is connected to it;
- The left-hand side of the diagram shows a physical representation of a LED, while the right-hand side shows the schematic symbol for a LED.



Led Light

16

- LEDs need to be connected the correct way around into a circuit, otherwise, they will not work;
- When looking closely at a LED, it is possible to notice a flat side on the LED casing;
- The leg on this side is the **cathode**, which connects to the **negative** or ground side of a power source;
- The cathode leg will also be the shorter of the LED's legs;
- The other leg is known as the **anode** and connects to the **positive** side of a power source.

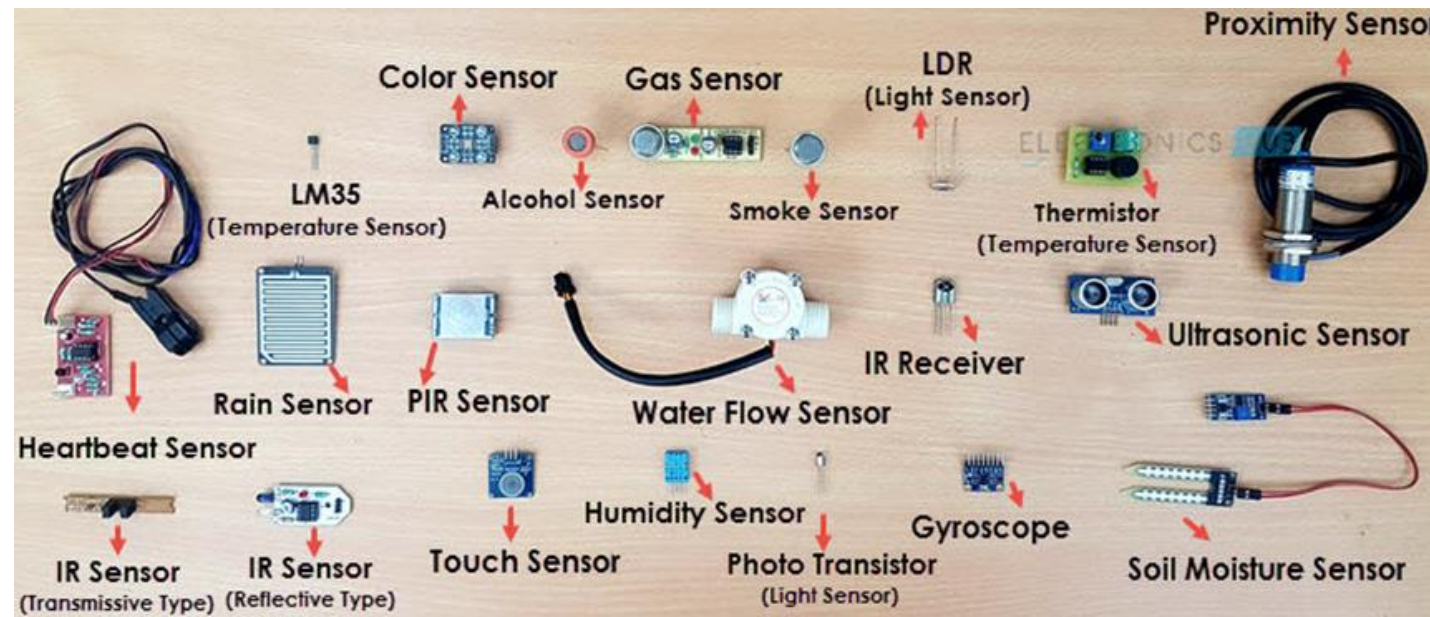
Sensors

Mobile Networks and Internet of Things

Sensors

18

- A sensor is a device that converts signals from one energy domain to an electrical domain;
- Sensors are usually a part of a bigger system, and can be analog or digital, according to the output they produce.



GPIO interface

Mobile Networks and Internet of Things

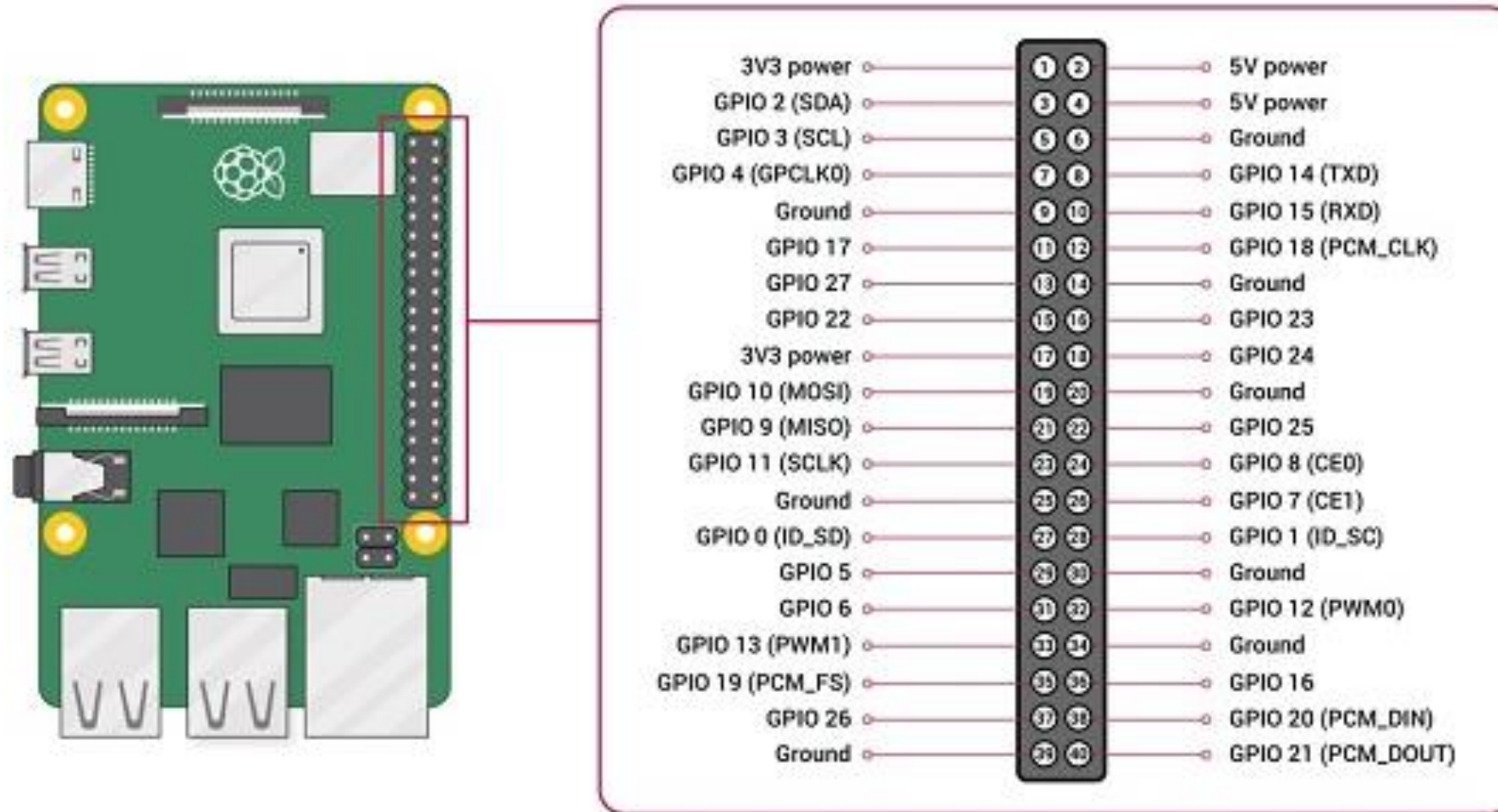
GPIO interface

20

- *GPIO = General-Purpose Input/Output;*
- Standard interface used to connect microcontrollers to other electronic devices;
- It can be used with sensors, diodes, displays, and System-on-Chip modules;
- The GPIO can be used in three modes:
 - Input: default, receive input from connected device (e.g., button);
 - Output: deliver data to connected device (e.g., led lamp);
 - UART interface: *Universal Asynchronous Receiver-Transmitter*, enables definition of custom advertising packets.

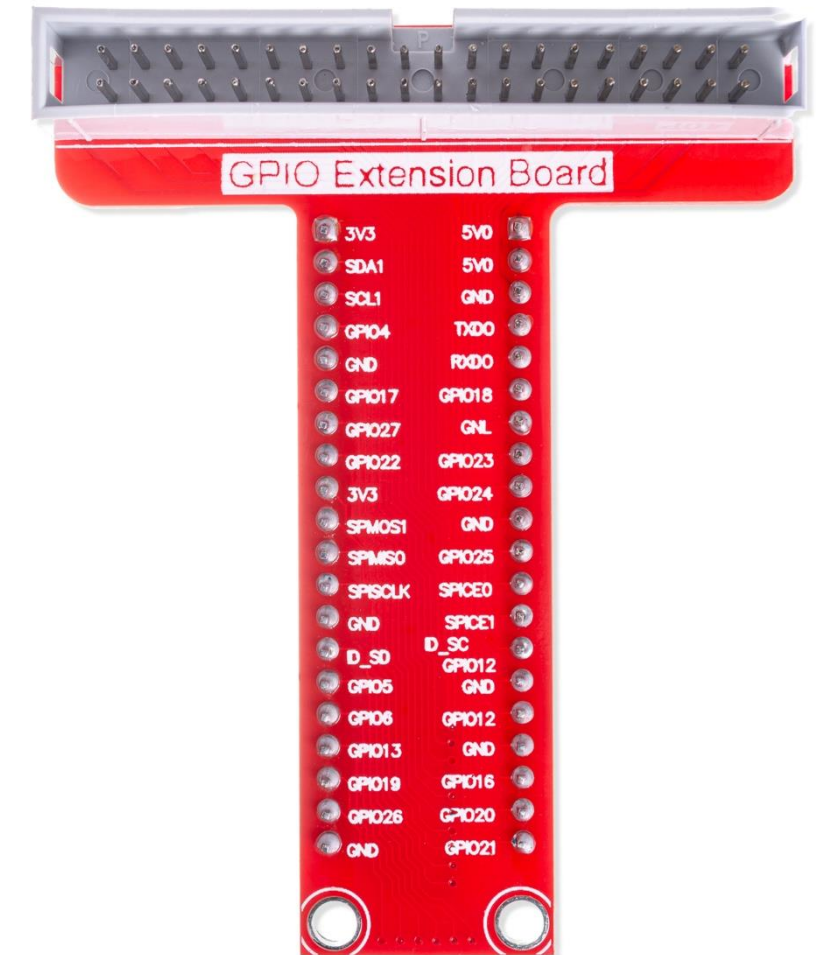
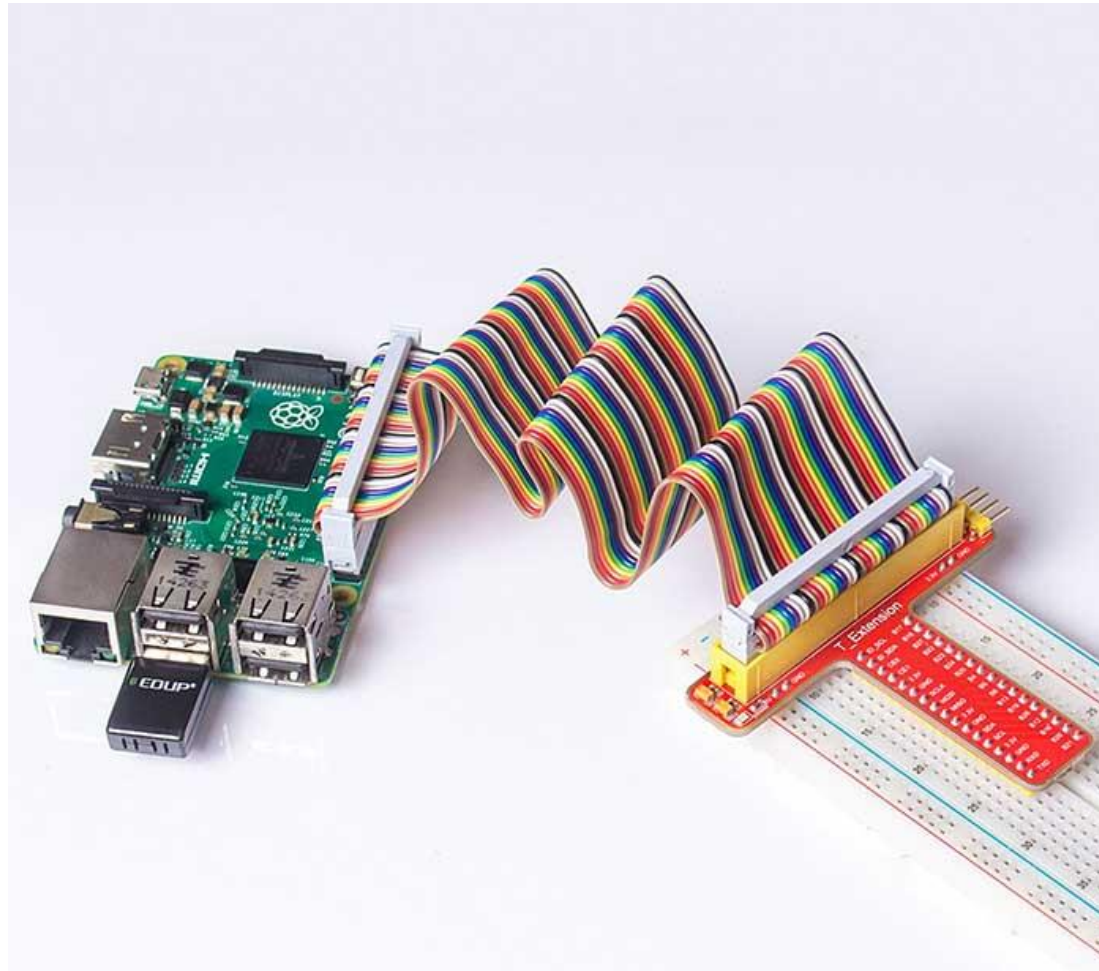
GPIO interface

21



GPIO interface

22



GPIO extension board

GPIO interface

23

- Voltages
 - There are two **5V pins**, two 3V3 pins (**3.3V**), and several ground pins (**0V**) – **these pins are not configurable!**
 - The **remaining** pins are general-purpose 3V3 pins, meaning that the outputs are set to 3.3 Volts and the inputs tolerate **at most** 3.3 Volts.
 - A GPIO pin designated as an **output pin** can be set to **high** (3.3V) or **low** (0V). Components attached to these pins will receive current if the output is set to high; otherwise, they will not receive current.
 - A GPIO pin that is designated as an **input pin** will allow a signal to be received by the Raspberry Pi. A voltage between 1.8V and 3.3V will be read by the Raspberry Pi as high; anything lower than 1.8V will be read as low.
 - Do not allow an input voltage above 3.3V, **or else you will damage your RPi!**

GPIO interface

24

- **Warning**
 - Do not attach 3V3 components directly to a 5V pin on the Raspberry Pi, or you may damage the component or your device
 - Certain components, such as LEDs, should have resistors to limit the current passing through them.

It is now time to get your Raspberry!

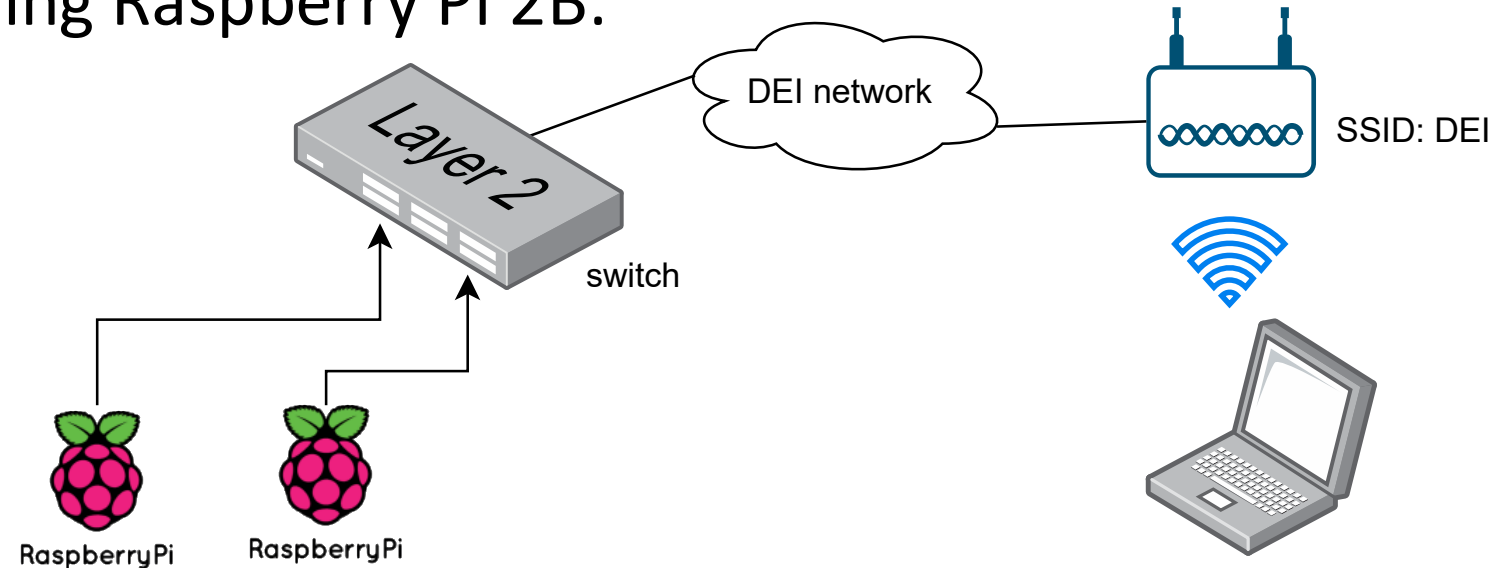
Class setup

Mobile Networks and Internet of Things

Class setup – using cable network

27

- Typical class setup using Raspberry Pi 2B:

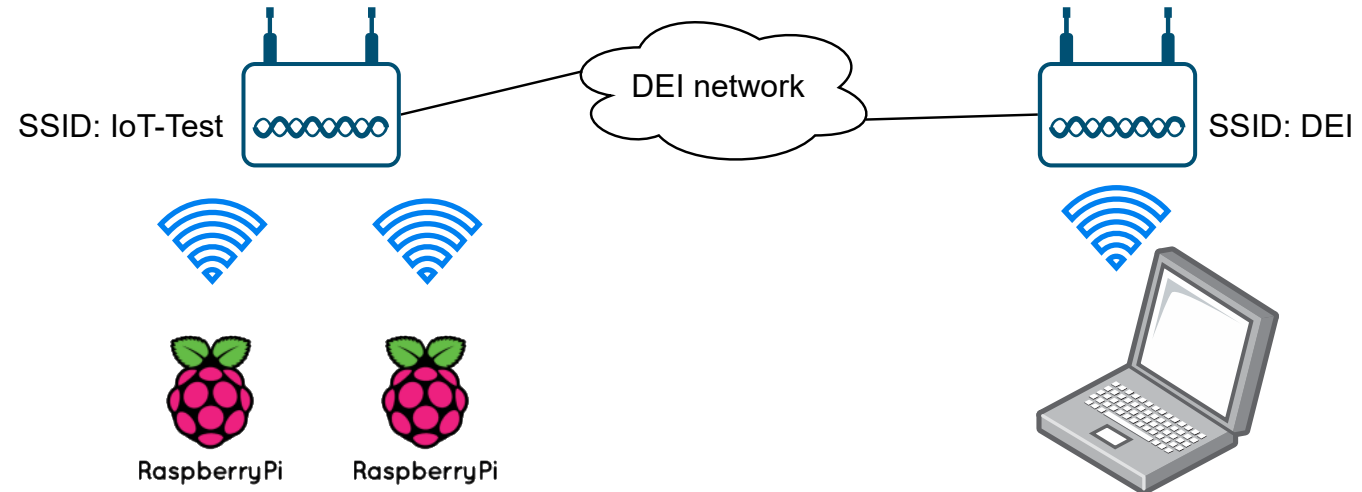


- Each group will work with one of the Raspberry Pi available. All of them are connected to a switch in the classroom and will be accessed using the Wi-Fi with SSID: **DEI** (eduroam will not work!).
- Make sure you are connected to the proper SSID before starting the assignment. The login/password of the “DEI” connection is the same as your mail.dei.uc.pt password

Class setup – using wireless network

28

- Typical class setup using Raspberry Pi Zero 2 WH:



- Each group will work with one of the Raspberry Pi Zero available. All of them are connected to the IoT-Test Wi-Fi network. They can be accessed using the Wi-Fi with SSID: **DEI** (eduroam will not work!).
- Make sure you are connected to the proper SSID before starting the assignment. The login/password of the “DEI” connection is the same as your mail.dei.uc.pt password

Raspberry Pi Access Info

Mobile Networks and Internet of Things

Raspberry Pi Access Info

30

- Access credentials to Raspberry Pi 2B devices:

- User: `user`
- Password: `password`

- IP addresses:

Raspberry Pi 2B	IP address
srsa-pi-1	10.6.1.2
srsa-pi-2	10.6.1.3
srsa-pi-3	10.6.1.4
srsa-pi-4	10.6.1.5
srsa-pi-5	10.6.1.6
srsa-pi-6	10.6.1.7
srsa-pi-7	10.6.1.8
srsa-pi-8	10.6.1.9
srsa-pi-9	10.6.1.10



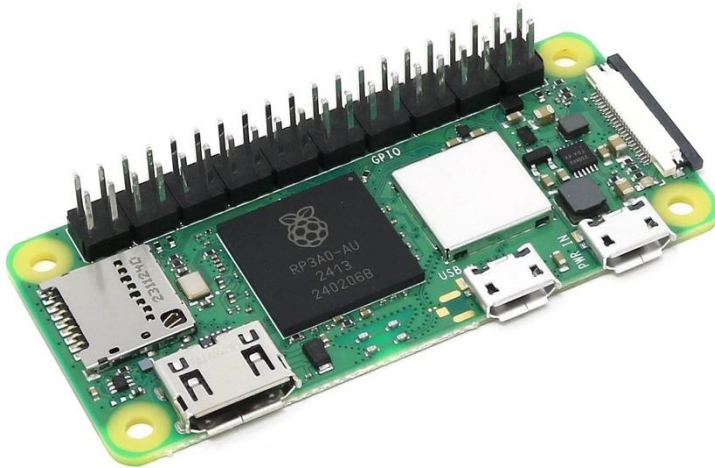
Raspberry Pi Access Info

31

- Access credentials to Raspberry Pi Zero 2 WH devices:

- User: `user` OR `rpiuser`
- Password: `password` OR `rpipass`

- IP addresses:



Raspberry Pi Zero 2 WH	IP address
srsa-pi-z-01	10.6.1.61
srsa-pi-z-02	10.6.1.62
srsa-pi-z-03	10.6.1.63
srsa-pi-z-04	10.6.1.64
srsa-pi-z-05	10.6.1.65
srsa-pi-z-06	10.6.1.66
srsa-pi-z-07	10.6.1.67
srsa-pi-z-08	10.6.1.68
srsa-pi-z-09	10.6.1.69
srsa-pi-z-10	10.6.1.70

Raspberry Pi Zero 2 WH	IP address
srsa-pi-z-11	10.6.1.71
srsa-pi-z-12	10.6.1.72
srsa-pi-z-13	10.6.1.73
srsa-pi-z-14	10.6.1.74
srsa-pi-z-15	10.6.1.75
srsa-pi-z-16	10.6.1.76
srsa-pi-z-17	10.6.1.77
srsa-pi-z-18	10.6.1.78
srsa-pi-z-19	10.6.1.79
srsa-pi-z-20	10.6.1.80

Accessing Raspberry Pi

Mobile Networks and Internet of Things

Accessing Raspberry Pi

33

- Connect to Wi-Fi with SSID = DEI
 - The username/password are the same of your *student.dei.uc.pt* account

- Check connectivity to your Raspberry Pi:

```
ping <ip_address>
```

E.g., `ping 10.6.1.9`

Accessing Raspberry Pi

34

- Connect to your Raspberry Pi using *ssh*:

```
ssh user@<ip_address>
```

E.g., `ssh user@10.6.1.9`

Students with
Windows systems
can use Putty

- After entering, check your directory using command *pwd*:

```
pwd
```

- The result should be like: `/home/user`

- Note:

- When using `ssh`, If "Host key verification failed" message appears, it means that the host key of the remote host was changed.
 - To remove it from your Linux/Mac PC, you can use: `ssh-keygen -R hostname`

Accessing Raspberry Pi

35

- From your base directory, create the directory where you will store your code:

```
$mkdir <your_directory>
```

- **Note:** Use a distinctive name in order to differentiate your directory from other students using the same device.
- Change to the new directory created:

```
$cd <your_directory>
```

Transfer files to the Raspberry Pi

Mobile Networks and Internet of Things

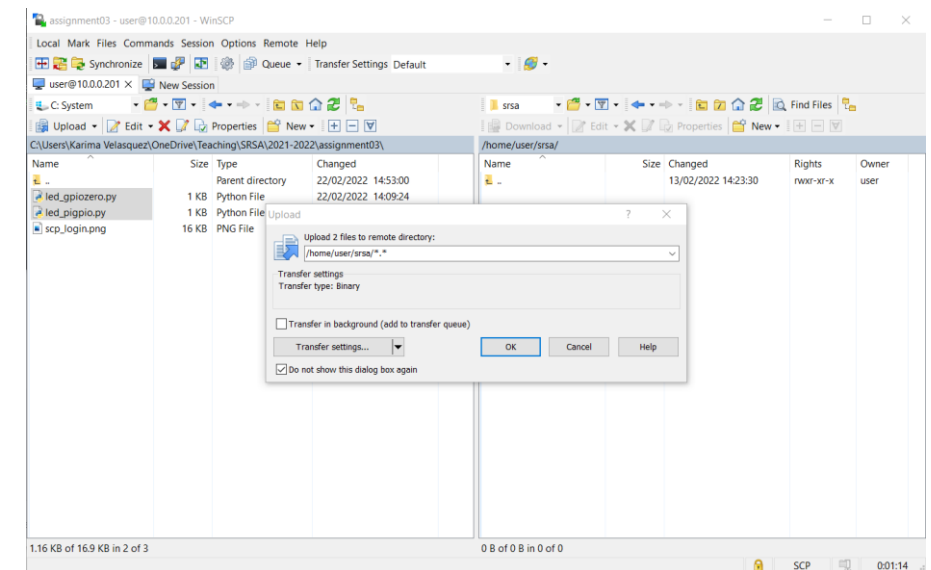
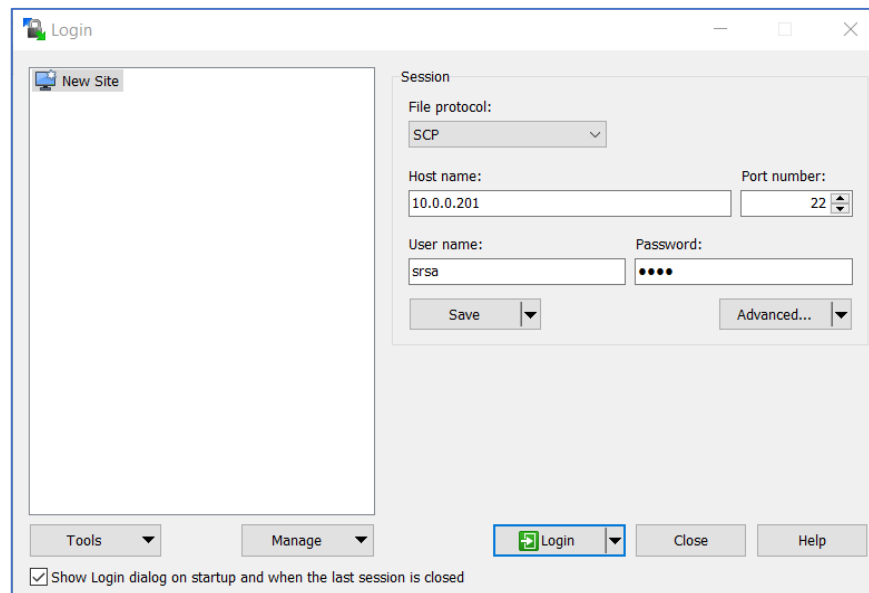
Transfer files to Raspberry Pi

37

- Send files from a MS Windows OS PC
 - Use the app **WinSCP** (<https://winscp.net/eng/download.php>) to transfer the files to your Raspberry Pi.

Fill the hostname field with the IP address or DNS name of your group Raspberry Pi; fill the username and password of that specific Raspberry Pi (will be given).

Select the *local* directory on the left, and the *remote* directory on the right. Then, drag and drop the files from the local to the remote directory. Then, confirm the transfer



Transfer files to Raspberry Pi

38

- Send files from a Linux based PC (or Mac)
 - Go to the command line.
 - Use command `scp` (Secure CoPy):

```
scp <file.py> <user@remote_address:/remote/directory/>
```

- Where:
 - user: is your login name
 - remote_address: the IP address or name of your Raspberry Pi
 - /remote/directory: the directory you want to access in the Raspberry

E.g.:

```
scp file.py user@10.254.0.1:/home/user/myfolder
```

- After the command, you will be asked to provide the password

Python Virtual Environments

Mobile Networks and Internet of Things

Python Virtual Environments

40

- The Python virtual environment is such that the Python interpreter, libraries and scripts installed into it are isolated from those installed in other virtual environments;
 - This enable sharing the same Raspberry between groups
- It contains a directory tree with Python executable files and other files, which indicate that it is a virtual environment;
- All Python related activity is sandboxed to your virtual environment.

Python Virtual Environments

41

- Execute the following command to create your environment:

```
$python -m venv <your_venv>
```

- This command creates a new Python virtual environment using the `venv` tool.
- The `-m venv` tells Python that we are going to run the `venv` module
- The `<your_venv>` parameter is the name of the folder where the virtual environment will be created.

Note: `python3` is used in some systems

Make sure you are using Python 3.x or some libraries might not work)

Python Virtual Environments

42

- Activate the virtual environment from your new directory:

```
$ source <your_venv>/bin/activate  
(venv) $
```

- When the terminal has a Python virtual environment activated, all Python related activity is sandboxed to it.
- If the Raspberry Pi is disconnected from power, after the Python virtual environment is created, it is not erased. However, it will have to be activated again to be used.
- To leave the virtual environment, type: `deactivate`

GPIO Zero library

Mobile Networks and Internet of Things

GPIO Zero library

44

- An entry-level and easy to use GPIO library for controlling simple electronics;
- Component interfaces are provided to allow a simple way to get started with physical computing;
- With very little code, you can quickly connect your components together;
- GPIO Zero builds on a number of underlying pin libraries, including RPi.GPIO and pigpio, each with their own benefits.

GPIO Zero library

45

- Use the pip command (Python Install Packages) to install the packages. As first step, upgrade the **pip** tool, using the following command:

```
(venv) $ pip install --upgrade pip
```

- This command might take some time to complete, and potentially output a lot of text to the Terminal;
- With pip upgraded, confirm which Python packages are already installed. For this, use the following command:

```
(venv) $ pip list
```

- **Note:** This commands must be executed inside the virtual environment



GPIO Zero library

46

- Install the GPIO packages using the pip install command as follows:

```
(venv) $ pip install gpiozero pigpio
```

- Execute the `pip list` command again and notice if there is any difference. Take a snapshot of the packages you have previously installed using the `pip freeze` command:

```
(venv) $ pip freeze > requirements.txt
```

- This command freezes all installed packages into a file named `requirements.txt`, which is a common filename to use for this purpose.



GPIO Zero library

47

- Look inside your `requirements.txt` file and notice its content.
You can use the following command:

```
(venv) $ cat requirements.txt
```



- This is a good practice in case if you move your Python project to another machine or a new virtual environment;
- You can use the `requirements.txt` file to install all required packages for your project to work;
- Notice that whenever you install new packages with `pip install` you also will need to re-run `pip freeze > requirements.txt` to capture new packages and their dependencies.

And now, some exercises...
Go to Assignment!